

Challenge: high-dimensional wake data

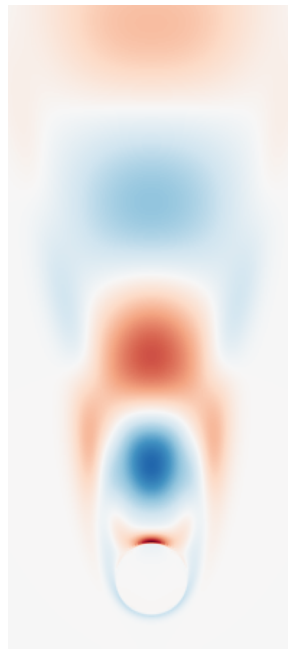


Thousands of spatial degrees of freedom make direct equation discovery difficult.



DMD

Need coherent latent modes

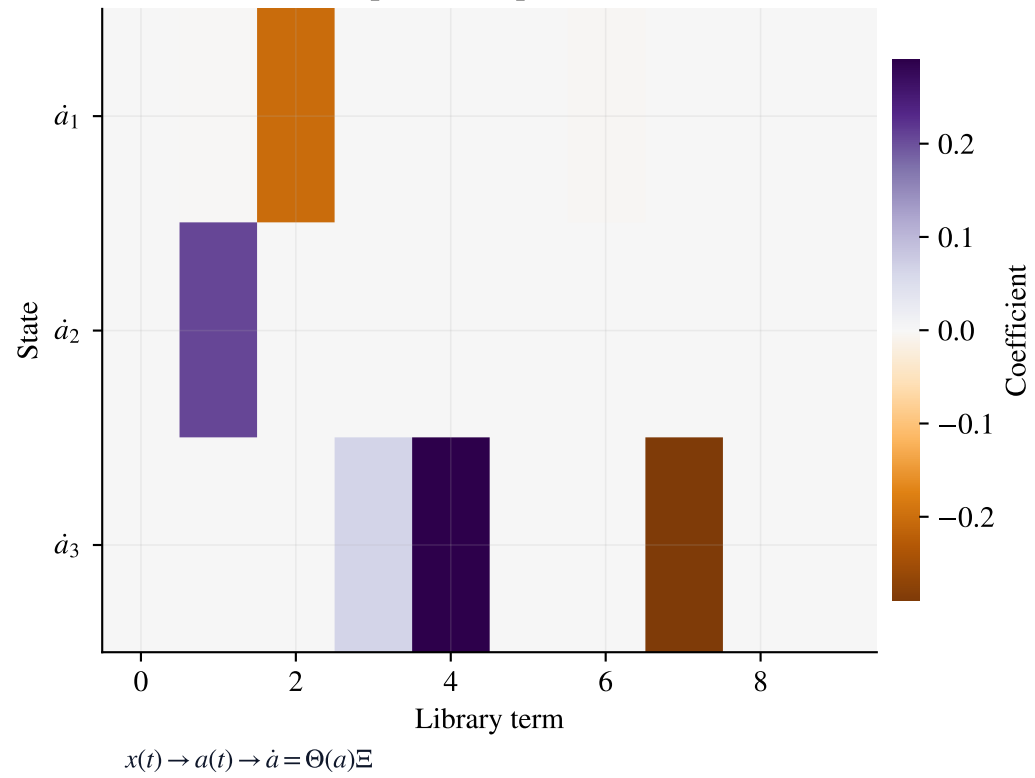


DMD extracts the dominant oscillatory structure and low-dimensional coordinates.



SINDy

Goal: sparse interpretable ODE



Use DMD for reduced coordinates and SINDy for compact governing-equation discovery.